

LangChain assistant

Day 2 — memory chat and direct tool demos

Memory chat **Tools**

Custom tools

Invoke each tool directly (no Bedrock required). Metadata comes from the LangChain `Tool` wrappers.

> Registered tools

Calculator

Perform basic math. Input: expression like '25 * 4'

Expression



100 / 4

Run calculator

CurrentTime

Get current time. Input: 'short', 'long', 'date', or 'default'

Format

long



Get time

WordCounter

Count words and characters. Input: text to analyze

Text

hello world

Count words

FileSearch

Search files in the current directory. Input: glob like '*.py' or substring like 'memory'

Pattern



*.py

Tool → chain (stretch)

Runs FileSearch first, then asks Bedrock to summarize using only that tool output.

Pattern for tool-result chain

Run FileSearch → summary chain

Tool result

```
Found 15 file(s) under /Users/stephenmckitrick/AICO-ECHO/langchain-assistant match
- langchain_chatbot_lab.py
- main.py
- scripts/run_day2_evidence.py
- src/__init__.py
- src/chains.py
- src/client.py
- src/memory.py
- src/prompts.py
- src/tools.py
- streamlit_app.py
- tests/__init__.py
- tests/test_chains.py
- tests/test_memory.py
- tests/test_prompts.py
- tests/test_tools.py
```

LLM summary (uses tool result)

The software project under `/Users/stephenmckitrick/AICO-ECHO/langchain-assistant` contains 15 Python files. The main components include the entry point in `main.py`, a Langchain chatbot script in `langchain_chatbot_lab.py`, and a Streamlit application in `streamlit_app.py`. The core logic is organized in the `src` directory, which includes modules such as `chains.py`, `client.py`, `memory.py`, `prompts.py`,

and `tools.py`. Additionally, the project includes an `scripts` directory with a script for running a specific day's evidence and a `tests` directory with test files for each module.